



Visualization of post-detonation fireball flowfields and comparison to CFD modeling

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ABSTRACT

Visualization of flow structures within post-detonation fireballs has been performed for benchmark validation of numerical simulations. Custom pressed PETN explosives with a 12-mm diameter hemispherical form factor were used to produce a spherically symmetric post-detonation flow with low soot yield. Hydroxyl-radical planar laser induced fluorescence (OH-PLIF) was employed to visualize the structure ranging from approximately 10 μ s to 35 μ s after shock breakout from the explosive pellet. Fireball simulations were performed using the HyBurn Computational Fluid Dynamics (CFD) package. Experimental OH-PLIF results were compared to synthetic OH-PLIF from post-processing of CFD simulations. From the comparison of experimental and synthetic OH-PLIF images, CFD is shown to replicate much of the flow structure observed in the experiments, revealing potential differences in turbulent length scales and OH kinetics. Results provide significant advancement in experimental resolution of these harsh turbulent combustion environments and validate physical models thereof.

1. Introduction

Detonation of a High Explosive (HE) releases an air-shock followed by rapidly expanding post-detonation products. The interface between the shock-heated air and fuel-rich products is hydrodynamically unstable, and the resultant mixing and oxidation forms a post-detonation fireball. Modeling and simulation of these dynamics are of interest to a variety of industrial and military applications.

Due to computational challenges stemming from coupled physics and the range of length- and time-scales, Computational Fluid Dynamic (CFD) predictions must solve simplified physics [1–3]. Experimental validation is therefore vital to prove the accuracy of these predictions. However, measurements are also challenged by the harsh environments and rapid transients.

Traditionally, model validation has relied on data from bomb calorimeters, pressure sensors, emission spectroscopy, and external high-speed imaging. For example, Kuhl and Reichenbach utilized pressure transducers within bomb calorimeters and calorimetric tunnels to observe post-detonation fireballs and after-burn of shock-dispersed fuel [1]. Fedina and Fureby performed simulations to estimate total heat release in comparison to experimental high-speed imaging and pressure transducer data [2]. Finally, the current team's prior work in Egel et al. performed CFD of a hemispherical explosive and showed good agreement between shock positions compared to

schlieren imaging and major product compositions from closed bomb gas analysis [3].

Diagnostics employed in these prior studies are incapable of quantifying the inter-dependent complexities of mixing and chemistry within the post-detonation fireball. To obtain such data, several authors have recently proposed *in situ* laser diagnostics [4–10]. Most have focused on absorption spectroscopy to measure thermodynamic variables and fireball opacities [4–8]. In Refs. [6,8], we use absorption data to validate temperatures and species from our CFD predictions. However, these comparisons are limited by the line of sight averaging inherent to absorption measurements.

To improve validation of turbulent mixing structures and dynamics, diagnostics are ultimately needed with fine spatial resolution. One possibility is Coherent Anti-Stokes Raman Scattering (CARS) as demonstrated by Richardson et al. [9]. However, CARS only provides data at a point or a small line, and impractical experimental repetition is needed to obtain the ensemble averaged flow. Alternatively, Gomez et al. quantified flowfield velocities using Particle Image Velocimetry (PIV) [10]. However, particle seeding challenges limit PIV to the shock-heated air.

Here, we propose unseeded Planar Laser Induced Fluorescence (PLIF) for 2D flow visualization. PLIF uses a spectrally tuned laser sheet to excite and image fluorescence in a molecule of interest, such as OH. OH-PLIF is commonly employed in gas-phase combustion

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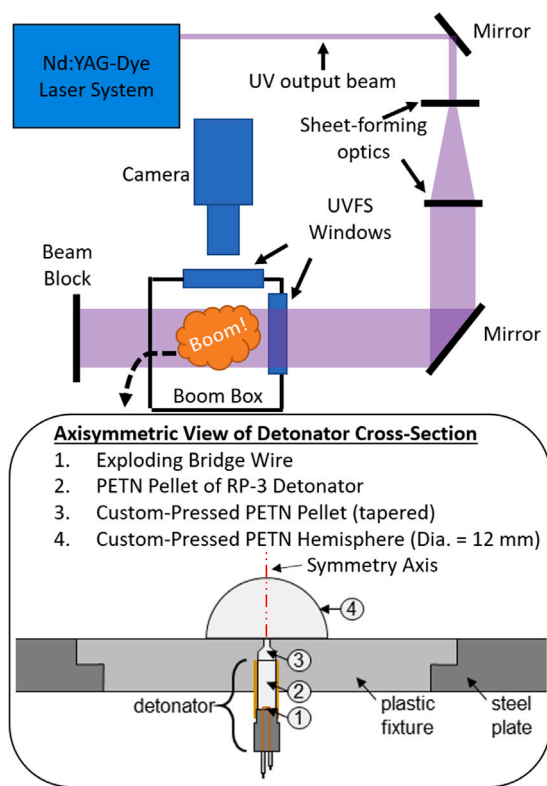


Fig. 1. Experimental setup utilized in the present study.

and detonation environments (see Mével et al. [11] for an example). However, prior to this work, OH-PLIF does not seem to have been applied to HE post-detonation flows. This lack of prior work likely stems from perceived measurement challenges caused by index of refraction gradients and signal absorption in the extreme environment. These challenges are confirmed in experiments presented here, and OH-PLIF modeling is proposed to interpret measurement results and enable model validation.

In summary, the objectives of this work are thus threefold: (1) demonstrate the first OH-PLIF diagnostics of reactive turbulent structures in post-detonation flows of a HE, (2) extend OH-PLIF modeling to enable comparisons with state-of-the-art reactive CFD predictions, and, (3) advance model validation using the developed tools.

2. Experimental setup

Post-detonation fireballs were generated using the configuration detailed in Fig. 1. Pentaerythritol tetranitrate (PETN) was pressed into 12 mm diameter hemispheres with a density of 1.60 ± 0.01 g/cc. Compared with other common HEs, PETN is relatively well oxygen balanced and produces little soot. Therefore, the PETN post-detonation flow field is relatively optically transparent, making it amenable to the PLIF diagnostics proposed here.

Pellets were initiated at their center via an assembly including a custom tapered PETN pellet and a commercial RP-3 detonator (Teledyne RISI). The additional custom PETN pellet was tapered to restrict the initiation spot size at the hemispherical pellet to roughly 1 mm. For each experiment, the HE was loaded onto a mounting plate such that the bottom of the hemispherical PETN pellet was flush with the plate. The HE assembly was then placed in a proof-tested enclosure, or boom box. All sides of this box were made of clear polycarbonate with UVFS windows incorporated into two adjacent sides, allowing a UV laser beam into the boom box and subsequent perpendicular emission of UV light from excited-state species pumped via the incoming beam.

A 10 Hz Nd:YAG laser (Spectra Physics Pro350) was frequency doubled to 532 nm and used to pump a Rhodamine dye laser (Sirah PrecisionScan) with a tunable output around 560 nm. This dye laser output was also frequency doubled with an output bandwidth of $\sim 0.05\text{--}0.1$ cm⁻¹. The UV output beam excited a rovibrational transition of the OH molecule present in the post-detonation fireball flow field. In this work, the UV radiation was generated at 282.75 nm corresponding to the Q₁(5) line of the (1,0) band of the OH A²Σ⁺-X²Π electronic transition. The laser pulse duration was 10 ns with a Gaussian shape and pulse energies were 12 mJ/pulse with sheet dimensions of approximately 1 mm × 30 mm, which resulted in fluences beyond the linear portion of the saturation regime. Fluorescence radiation was collected with a 16-bit, 4-megapixel, intensified sCMOS camera (PCO Dicam C1) utilizing a 100-mm f/2.8 compound UV lens (Cercor) with a Semrock notch filter centered at 315 nm (± 15 -nm FWHM). The center wavelength of the Semrock filter was selected to avoid scattering of the excitation source laser by only collecting off-resonance fluorescence. Furthermore, a low-pass filter with a cutoff wavelength near 400 nm reduced signal from background thermal emission as the camera was sensitive at such wavelengths and the Semrock filter did not fully attenuate IR radiation in this range. The camera gate was set at 10 ns with the gate delay determined by maximizing the LIF signal during visualization of a H₂-O₂ flame from a Hencken burner. A Shimadzu HPV-X2 camera equipped with a 640 nm filter (± 50 nm FWHM) was placed opposite the PCO camera for high-speed imaging of the fireball (not illustrated in Fig. 1).

The PLIF camera and laser pulse were synchronized to record images of the OH flow field ranging from 10–35 μs after initiating the HE. The 10 Hz laser pulse rate limited camera acquisitions to one image per experiment (single-shot imaging). Consequently, specific flow structures visible in each OH-PLIF image are unique and cannot be correlated between experiments. Still, global features, such as shock locations, are highly repeatable between shots. Lastly, experimental times reported herein are defined as the time after shock breakout from the PETN hemisphere.

Acquiring two or more images within the 35 μs experimental duration would require acquisition rates of at least 50 kHz. This is only recently feasible at significantly higher complexity and cost using pulse-burst laser technology [12]. In addition, pixel and noise limitations of high-speed imagers significantly degrade data quality. Because of these disadvantages, we specifically chose not to attempt high-speed OH-PLIF videography in this work.

Example OH-PLIF experimental images are provided in Fig. 2. It was desirable to verify that the location of observable features were spatially and temporally aligned between various data streams without manipulating the timing artificially between simulation and experiment. For this, the simulated inner and outer shock locations are imposed on the images giving confidence that the simulations align spatially and temporally with experiments, with more validation shown via schlieren imaging later. Additionally, a best-effort attempt at manually fitting a circle to the inner curvature of the OH layer was performed for images wherein this feature was recognizable. This was the case for Fig. 2(a), but was not possible for images like frames (b) and (c) of Fig. 2.

3. Simulation setup

Simulations of the PETN detonation, shock breakout, and fireball formation were performed using the HyBurn code developed at the University of Florida. A full discussion of the model is provided in Egel et al. [3]. The governing equations have been included in the supplementary material for the convenience of the reader. The reactive Euler equations were solved in 3D Cartesian coordinates. Initial conditions consisted of a 12 mm diameter center-initiated hemispherical PETN charge in ambient air at 298 K and 1 bar with an initial charge density of 1.6 g/cc. The domain was 0.3 m × 0.3 m × 0.3 m. Two

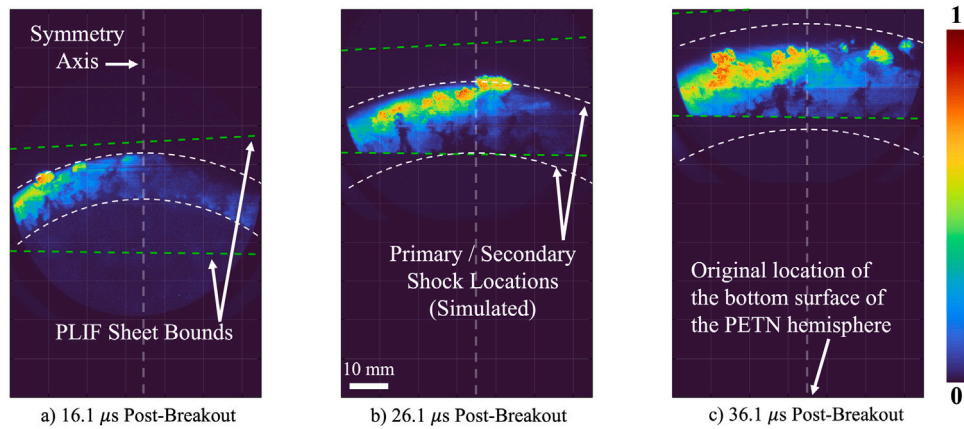


Fig. 2. Experimental PLIF images after shock breakout from the hemispherical charge. As the laser sheet traverses the fireball (left-right), PLIF signal intensities display asymmetry from strong laser absorption in the dense environment. Also, the flow appears more asymmetric at later times as a smaller total portion of the fireball is imaged. At these times, the overall shape and curvature of the fireball is not visible but rather the more stochastic behavior of local eddies.

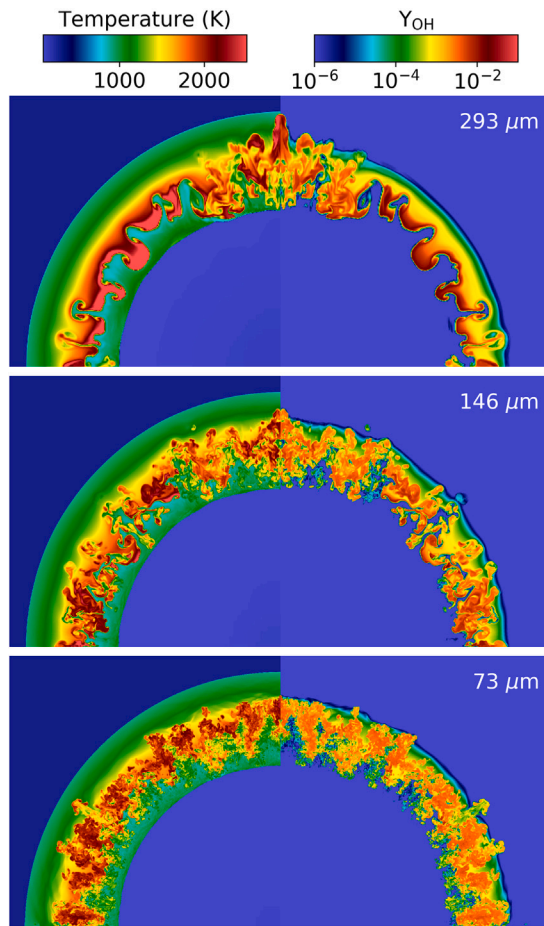


Fig. 3. Slice plots of temperature and OH mass fraction for simulations with minimum grid sizes of 293, 146, and 73 μm at 50 μs .

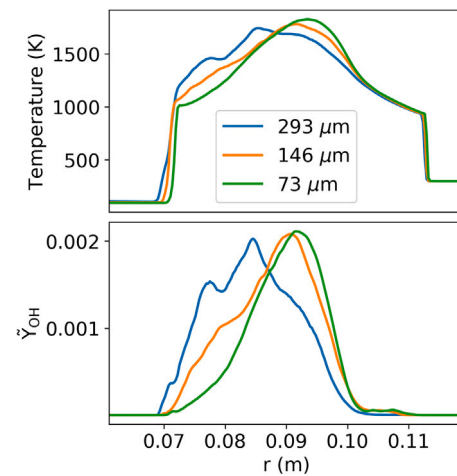


Fig. 4. Azimuthal averages of temperature and OH mass fraction for simulations with minimum grid sizes of 293, 146, and 73 μm at 50 μs .

simulations were performed. The first utilized dry air consisting of O_2 and N_2 as the ambient gas-phase mixture, while the other included an additional 1% mass fraction of H_2O in the ambient air, hereafter referred to as the dry and water simulations, respectively. The detonation is progressed through the charge using a simplified programmed burn model [13] that converts the solid explosive into hot, dense detonation product gases at the Chapman–Jouguet (CJ) state provided by the Tiger equilibrium code [14,15]. The equation of state for the gas-phase is

the Becker–Kistiakowsky–Wilson equation of state [16] and a skeletal chemical reactions mechanism is used for the gas-phase reactions [17].

A grid convergence study was performed in 3D Cartesian coordinates with a 1% mass fraction of H_2O in the ambient air. The base grid was 0.3 m in the x-, y-, and z-directions with symmetry boundary conditions on the lower end and outflow boundary conditions on the upper end. The simulations used adaptive mesh refinement (AMR) with 3, 4, and 5 maximum levels of refinement resulting in minimum grid sizes of 293, 146, and 73 μm , respectively.

Comparisons between all three grid sizes are shown in Fig. 3. Slice plots of temperature and OH mass fraction were taken at 50 μs . Successive refinement results in more detailed vortical structures and increased mixing between the detonation products and air. However, all three simulations show qualitatively the same bulk flow. The locations of the primary and return shocks are identical along with the approximate location of the OH layer formed by air dissociation.

Average values were calculated with an azimuthal averaging process detailed in [18,19]. The average temperature and OH mass fraction at 50 μs are shown in Fig. 4. The temperature was averaged with volume and the mass fraction was Favre averaged. The qualitative and averaged value comparisons in Figs. 3 and 4 show convergence with increasing grid refinement. Simulations with grid sizes of 146 μm and 73 μm behave similar enough for the current study. Results that follow correspond to the finest grid.

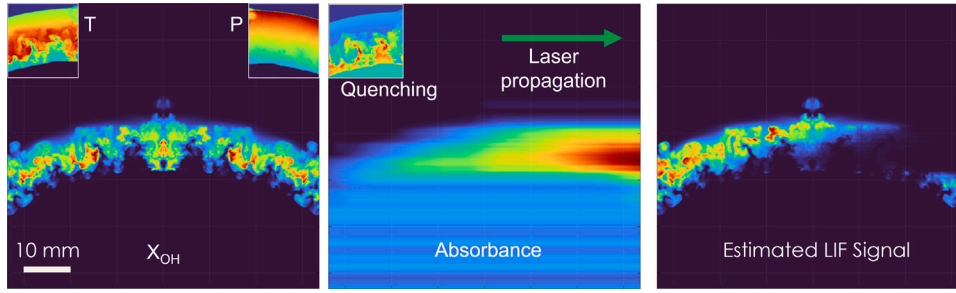


Fig. 5. Simulated flowfield variables from a CFD simulation with ambient water (left) demonstrating the implementation of the spectral model (middle) to generate a synthetic PLIF signal (right).

4. OH-PLIF signal estimation: Spectral modeling

The calculation of a synthetic OH-PLIF signal as estimated from the CFD simulation was performed for comparison of experimental OH-PLIF data. Prior efforts related to calculating absolute fluorescence levels, such as that by Xiong et al. [20], have proposed multilevel collisional models for predicting signal levels as a function of OH concentration. Such models are able to capture collisional interactions occurring during and after the excitation pulse. These include saturation effects dependent on laser fluence, ro-vibrational energy transfer to equilibrate the highly non-Boltzmann upper-state energy distribution after excitation, and the accompanying time-dependent decay to the various ro-vibrational ground state levels from either spontaneous emission or quenching. Here, we instead use a simplified variant, which allows for qualitative spatial comparison of flow features but not absolute fluorescence levels.

The simplified model [21] is given by Eq. (1). The primary assumption of this 2-level model is that fluorescence is emitted at the same wavelength as that of excitation: OH is excited to an upper state and decays spontaneously, or is quenched, back to the ground state. In calculating the saturated LIF signal, $S_{f,sat}$, the only direct input from the simulation is the ground-state OH number density, n_1^0 , where V is the local grid point volume, A and B are Einstein transition probabilities, and Ω is the solid angle of collection. Additional inputs from the simulation (T , P , collider species concentrations), are incorporated to calculate the quenching rate, Q_{21} , using thermally-averaged quenching coefficients. Quenching coefficients are taken from Tamura et al. [22], which includes the following species as colliders: OH, H, H₂, H₂O, N₂, NO, CO, CO₂, and O₂. In general, the most influential collider was found to be H₂O, followed by CO₂, although this is dependent upon local thermochemical conditions and not identical over the entire flow field. The relation between the transition from linear to saturated fluorescence regimes, $S_f/S_{f,sat}$, accounts for laser fluence levels, I_v , at each grid point. The remaining variables are dependent upon the input laser power, laser linewidth, $\Delta\nu_{laser}$, and grid size to calculate the laser fluence.

$$S_{f,sat} = n_1^0 V \left(\frac{B_{21}}{B_{12} + B_{21}} \right) A_{21} \frac{\Omega}{4\pi} \quad (1a)$$

$$\frac{S_f}{S_{f,sat}} = \frac{1}{1 + \frac{Q_{21} + A_{21}}{(B_{12} + B_{21})I_v}} \quad (1b)$$

$$I_v = \frac{Power \cdot \tau}{Area \cdot \Delta\nu_{laser}} = \frac{Fluence}{Pulse} \text{ (J/m}^2\text{)} \quad (1c)$$

$$\alpha = -\ln(\tau) = -\ln\left(\frac{I}{I_0}\right) = n_1^0 \sigma_v L \quad (1d)$$

Local variations in laser fluence were accounted for by calculating a synthetic absorbance of the laser through the CFD flow field with a 3-line absorption model using Sandia's SPEARS API [23]. Local laser absorbance, α (or transmission, τ), was calculated from CFD-supplied variables of ground state OH number density, n_1^0 , temperature, T , and

cell width, L , by using the well-known Beer–Lambert relation. Spectral parameters for calculating the temperature-dependent absorption cross-section, σ_v , were taken from ExoMol [24]. The lines selected for calculating absorption were the excitation line of $Q_1(5)$, along with the $Q_{21}(5)$ (not to be confused with quenching rate, Q_{21} , discussed previously) and $R_{21}(14)$ lines. Other lines could have been included, but this would have been unnecessary as the linewidth of the laser was narrow enough and distance in wavelength space from other transitions great enough that negligible amounts of absorption of the PLIF laser were expected from other lines.

An example of the data and process of generating synthetic LIF images is given in Fig. 5. The left frame includes a 2-D slice of normalized OH mole fraction, along with cropped insets of the temperature and pressure fields. Absorbance of the laser through the flow field in the center frame (with calculated normalized quenching rates inset in the upper left of the frame) correlates to the behavior of the resulting synthetic LIF signal in the right-most frame of the image. As the laser is absorbed from left-to-right through the flow field, the LIF signal levels decay. For the calculations performed herein, no deviation in the shape of the PLIF sheet was incorporated. While the PLIF sheet is expanding vertically from left-to-right in the experiments, the resulting change in local fluence levels was not expected to alter the interpretation of the data. Additionally, no localized distortion of the PLIF sheet was considered. Finally, density gradients present in the flow may have generated regions in which refractive index varied enough to cause thermal lensing of the PLIF sheet. Although possible, such phenomena were not observed to affect the experimental data or necessitate inclusion of these physics in the synthetic LIF signal scheme.

5. Experimental results and model comparisons

Comparisons between experimental data and simulation data are presented here. Simulation results were averaged over a 1 mm thick rectangle centered on a slice through the center of the charge, taking into account the signal averaging from the 1-mm thick laser sheet used in the experiments.

A campaign of schlieren imaging was performed to provide confidence that the inner and outer shock locations were in agreement between modeling and simulation. An in-house algorithm utilizing the computation of Sobel indices was written to track the location of the primary shock in processing the schlieren data. Estimation of the inner shock layer location was also performed wherein 5 points along the shock were user-selected and averaged with an estimated uncertainty of approximately ± 2.5 mm. This data validated that the timing of the OH-PLIF and schlieren experiments were in agreement relative to shock breakout.

A shock radius vs time plot for inner and outer shocks is given in Fig. 6 for the data collected in the present work. Simulated shock radii and experimental shock-tracking data for post-breakout times up to approximately 35 μ s agree within the uncertainty of the schlieren data for both the inner and outer shocks. Such agreement provides confidence that the energy release, and subsequent shock trajectory, modeled by

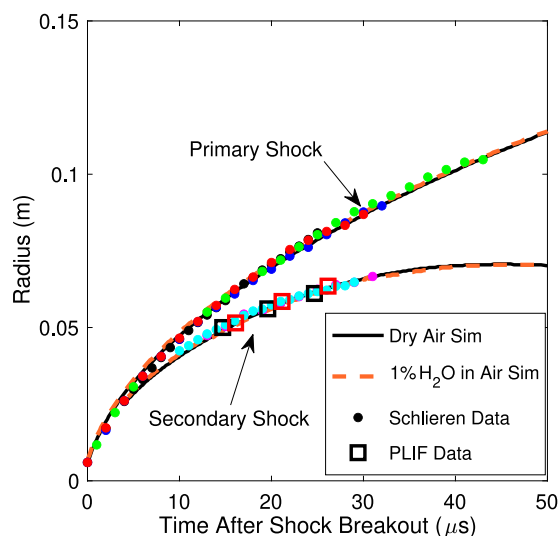


Fig. 6. Experimental shock radii as a function of time after shock breakout. Estimated uncertainties in radii are ± 2 mm and temporally at ± 1 μ s. Schlieren data points are color-coded for individual experiments. PLIF data are colored according to the experimental campaign. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

HyBurn agrees well with the experiment at the times investigated. Additionally, for OH-PLIF data wherein inner shock curvatures were not cutoff by the edges of the PLIF sheet (as in Fig. 2a), inner shock radii agreed well with that of the schlieren data. Such agreement confirmed proper temporal anchoring of experimental OH-PLIF for comparison with simulated OH-PLIF.

An example comparison of synthetic and experimental OH-PLIF data from the PLIF campaign is provided in Fig. 7. From the experimental OH-PLIF data, the reactive fireball layer is clearly visible between the inner shock and outer shock for the times investigated herein, bounding the reactive portion of the fireball. Absorption of the PLIF sheet from left-to-right is also clearly observed with the strongest PLIF signals coming from the regions of the flow field where laser fluence was highest, which is captured well by the synthetic PLIF calculations. Such strong absorption is not commonly observed in atmospheric or low pressure flames but has been previously reported for gas-phase detonations [11]. As shown here, simulations of this effect are vital for accurate interpretation of the experimental images.

Initial simulations were performed with no water in the pre-shock air mixture. This is shown in Fig. 7c. Fluid mixing is visible as evidenced by eddies resulting from Rayleigh–Taylor instabilities at the contact surface of the detonation products and air. Unlike the experiments, the dry-air simulation shows only well-defined structures through the reactive layer of the fireball and no background OH field. These differences motivated a second simulation including 1% mass fraction of H_2O in the ambient air mixture. The leading shock dissociates ambient H_2O , generating a region of OH between the leading shock and the contact surface of the detonation products and air, as observed in Fig. 7a.

6. Discussion

In combustion research, PLIF is a workhorse diagnostic for a wide variety of reactive environments. But until now, its utility has yet to be realized in HE fireballs. As shown by the OH-PLIF data herein, the new application of OH-PLIF to HE fireballs enhances flow field visualization by providing data previously inaccessible with conventional HE fireball diagnostics. In particular, the ability to resolve turbulent structures and length scales is significantly improved compared to high speed imaging

and schlieren. This new data is useful not only to elucidate the internals of the fireball, but also to assess modeling results.

Comparison of the turbulent mixing structures shows that both simulations and experiments exhibit large eddy structures on the order of 5–10 mm along with a cascade of finer scale eddies. As seen in the grid refinement results of Fig. 3, the finest turbulent dissipation scales are not yet captured due to computational limitations. Still, experiments and simulation comparisons shown in Fig. 7 are a significant advancement over the prior state-of-the-art that largely relied on line-of-sight schlieren. It is therefore anticipated that the OH-PLIF methodology proposed here will enable future efforts to improve turbulence modeling in extreme post-detonation flow fields.

In addition to these hydrodynamic considerations, OH-PLIF also improves our validation of reaction chemistry. In the experimental OH-PLIF images, the leading shock appears to dissociate some of the ambient H_2O to generate a background OH concentration, which is separate from that created by the detonation products. This observation motivated us to perform simulations with both dry and wet air. Indeed, the ambient water simulations confirm the existence of OH between the leading shock and the detonation product contact surface (Fig. 7a). This has not been previously reported in the post-detonation literature. Incorporation of this effect in future simulations will improve prediction of OH dynamics and is likely important for interpretation of other validation metrics, such as H_2O concentrations from line of sight absorption.

As shown in Fig. 2, the experimentally observed edge of the reactive OH layer aligns with leading air shock (within uncertainties) up to and including post-breakout times of roughly 26 μ s. However, in simulations, the outer edge of the OH layer appears to diverge from the shock-front. These differences are detailed further in Fig. 8. Ambient humidity levels were not recorded at the time of the experimental shots, and simulations were performed with a representative 1% H_2O mass fraction. It is hypothesized that the current differences in the apparent OH front are caused by the combination of poorly known vapor concentration and signal detection limits. To explore this further, future experiments should record ambient water vapor concentration. In addition, now that PLIF has been demonstrated feasible in this environment, other molecules of interest might also be considered. CH-PLIF, for example, may enable visualization of just the carbon-containing post-detonation products.

Finally, as a guide for future investigators, lessons learned from this work are summarized here. As mentioned previously, a low-sooting explosive formulation was necessary to avoid blockage of the PLIF sheet by soot. Despite this, the fireball was still very dense, particularly at early times. Thus, a laser with narrow linewidth and high pulse energy was needed to penetrate the fireball and visualize OH-PLIF over the entire span. Second, shock tracking with schlieren provided critical temporal anchoring of the recorded PLIF images. This highlights the need for a multi-pronged approach in selecting diagnostics to provide meaningful data for validating simulations while also being able to probe the harsh transient environment of HE fireballs. Lastly, recording the ambient humidity level is necessary to accurately simulate water dissociation by the leading shock, as demonstrated by the differing results of simulations with and without ambient water.

7. Conclusion

Post-detonation fireballs from custom HE pellets were visualized with OH-PLIF and schlieren imaging. With these *in situ* techniques, flow structures within the fireball are clearly visualized, enabling comparison to numerical simulations. The schlieren results provided temporal anchoring of PLIF data and simulations. Temporal agreement of shock radii between simulation and experiments implies good agreement of the energy release predicted by simulations. Furthermore, the simulations were able to replicate the larger spatial scaling of turbulent

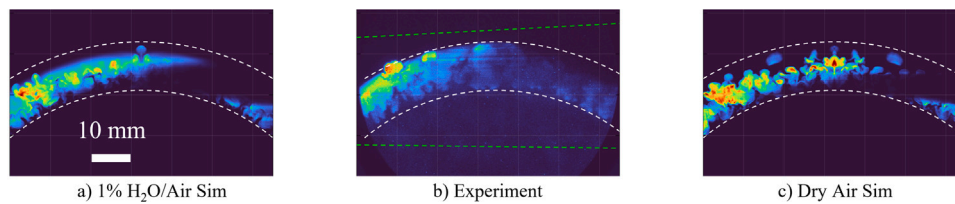


Fig. 7. Comparison of experimental data (b) with simulations performed in air with 1% mass fraction H₂O (a), and dry air (c), at $\sim 16.1 \mu\text{s}$ after shock breakout. Simulated shock locations are shown as white dashed lines.

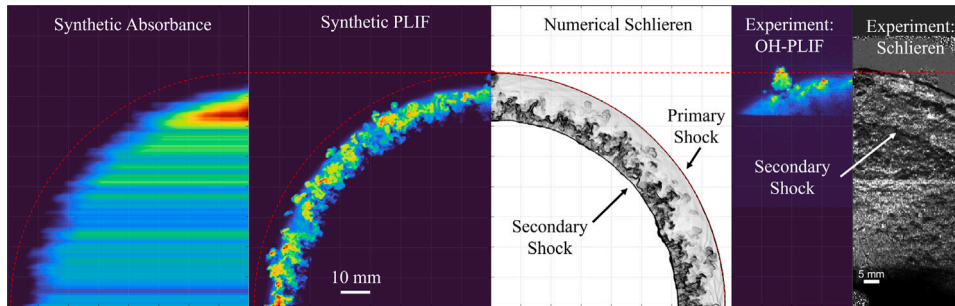


Fig. 8. Comparisons (left-to-right) of synthetic laser absorbance, synthetic OH-PLIF signal, numerical schlieren, experimental OH-PLIF signal, and experimental schlieren 24.5 μs post-breakout of the leading shock. The leading edge of the experimental OH-PLIF field aligns with the leading shock in the experiment while the leading edge of both the synthetic absorbance and synthetic OH-PLIF fields do not align with the simulated leading shock.

flow structures within the post-detonation fireball at the times investigated experimentally. There are also qualitative discrepancies in the fine spatial scales of the turbulent structures and location of the OH leading edge in comparison to experimental OH-PLIF images, along with differences in local spatial gradients of the observed OH-PLIF signal within the flow. Prior to this work, such differences between experiments and simulations could not be resolved. In summary, the OH-PLIF diagnostic utilized herein has provided meaningful data for HE model advancement.

CRedit authorship contribution statement

Joshua W. Hargis: Designed experimental research, Performed experiments, Analyzed data, Wrote the paper. **Anthony Egel:** Performed CFD simulations, Analyzed data, Wrote the paper. **Ryan Houim:** Advised implementation of CFD simulation, Wrote the paper. **Daniel R. Guildenbecher:** Designed experimental research, Performed experiments, Analyzed data, Wrote the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.proci.2024.105230>.

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